

HOUSE PRICE PREDICTION – SUMMARY REPORT

DATASET

Records : 545 rows × 14 columns
Price range: ₹1.75M – ₹13.30M
Mean price : ₹4.77M | Median: ₹4.34M

MODEL PERFORMANCE

Metric	Linear Regression	Random Forest
MAE	₹ 970,043	₹ 1,014,947
RMSE	₹ 1,324,507	₹ 1,399,769
R ² Score	0.6529	0.6124
Winner →	Linear Regression	

TOP FEATURES (Random Forest Importance)

1. area	0.468 (46.8%)
2. bathrooms	0.153 (15.3%)
3. air conditioning	0.060 (6.0%)
4. parking	0.058 (5.8%)
5. stories	0.057 (5.7%)
6. bedrooms	0.048 (4.8%)
7. furnishing_status_unfurnished	0.037 (3.7%)
8. basement	0.032 (3.2%)
9. prefarea	0.028 (2.8%)
10.hotwaterheating	0.018 (1.8%)
11. guestroom	0.018 (1.8%)
12.furnishingstatus_semi-furnished	0.013 (1.3%)
13.mainroad	0.010 (1.0%)

INSIGHTS

- Q1. Which features influence price most?
Area is the #1 driver (~47%), followed by bathrooms (~15%), AC, parking, and stories. These 5 explain 80%+ of price variation.
- Q2. How accurate was the model?
Linear Regression explained 65.3% of price variation ($R^2=0.653$) with an average error of ₹9.7 lakh per prediction.
- Q3. What surprised you?
Bedrooms matter LESS than bathrooms – buyers value amenities over room count. Hot water heating also signals luxury strongly. Prices are right-skewed with a long luxury tail above ₹9M.
- Q4. Business recommendation:
Price listings using area + bathrooms + AC + parking + stories. For sellers renovating, adding a bathroom or AC gives the best ROI. Feature these prominently in listings over bedroom count.